

**Part A: Negative Number Operations***Evaluate each expression completely.*

1.  $-10 + (-11)$

2.  $-2 + 6$

3.  $-4 - 4$

4.  $-7 - (-4)$

5.  $-1 + (-4)$

6.  $-8 \cdot 2$

7.  $-20 \div 4$

8.  $-36 \div 6$

9.  $-6 \cdot (-4)$

10.  $-9 \cdot (-2)$

11.  $-19 + (-2) \cdot (-4)$

12.  $21 \div (-3) - 5$

13.  $12 + (-10) \cdot (-6)$

14.  $-120 \div 12 - 2$

15.  $5 + 10 \cdot (-12)$

16.  $(-2)^2 - (-18) \cdot 3$

17.  $(-9)^2 - 9 \cdot 2$

18.  $19 - 2 \cdot (5 + 19)$

19.  $-19 - 5 \cdot (2 + (-19))$

20.  $(3 + (-6))^2 \div (-1)$

## Part B: Solving Linear Equations

*Solve each equation for  $x$ . Show your steps.*

1.  $2x = -18$

2.  $3x = -12$

3.  $x + 1 = 6$

4.  $3x = 15$

5.  $x - 8 = -16$

6.  $5x - 2 = 28$

7.  $4x - 7 = 9$

8.  $7x + 6 = -29$

9.  $3x - 4 = 8$

10.  $8x - 5 = -37$

11.  $9(3x + 12) + 12 = 93$

12.  $6(5x - 6) + 9 = -237$

13.  $7(8x - 6) - 9 = 341$

14.  $8(3x + 12) + 4 = 268$

15.  $4(7x - 11) - 10 = -306$

16.  $-x - 7 = 7x + 73$

17.  $-x + 4 = 3x + 44$

18.  $5x - 12 = 2x - 1$

19.  $-11x + 5 = 8x + 195$

20.  $8x + 2 = 12x + 14$

**Part C: Linear Inequalities***Solve each inequality for  $x$ .*

1.  $x - 9 \geq 2$

2.  $x - 6 \geq -6$

3.  $x + 1 \leq -4$

4.  $-4x \geq -16$

5.  $3x > 9$

6.  $x + 6 > -2$

7.  $x - 2 \geq 9$

8.  $8x + 9 < 41$

9.  $-8x + 8 < -8$

10.  $2x - 4 < -18$

11.  $-9x + 4 \leq -50$

12.  $-8x + 5 \leq -27$

13.  $2x + 3 \leq 19$

14.  $9x - 1 < 8$

15.  $10x - 11 < 8x - 27$

16.  $3x - 9 \leq 9x + 51$

17.  $-4x + 10 > 4x + 50$

18.  $-7x - 9 \leq -8x - 14$

19.  $10x - 9 \geq -6x + 151$

20.  $9x - 10 < 4x - 50$

## Part D: Exponent Rules

*Simplify. Write answers with positive exponents.*

1.  $x^4 \cdot 2x^6$

2.  $2x^3 \cdot 3x^2$

3.  $x^6 \cdot 2x^2$

4.  $4x^2 \cdot x^2$

5.  $\frac{4x^6}{x^2}$

6.  $\frac{2x^8}{x^4}$

7.  $\frac{4x^{12}}{x^5}$

8.  $\frac{4x^7}{x^2}$

9.  $(2x^8)^2$

10.  $(2x^9)^2$

11.  $(5x^2)^2$

12.  $(8x^4)^4$

13.  $7 \cdot (4^2)^{-2}$

14.  $8 \cdot 3^{-4}$

15.  $\frac{8}{5^{-4}}$

16.  $\frac{2}{7^{-1}}$

17.  $7^4 \cdot 7^{-1}$

18.  $3^2 \cdot 3^0$

19.  $3^4 \cdot 3^{-1}$

20.  $7^2 \cdot 7^0$

## Part E: Slope and Linear Graphs

*Identify the slope  $m$  and the  $y$ -intercept  $b$  of each line.*

1.  $y = -6$

2.  $y = x + 6$

3.  $y = 2x - 3$

4.  $y = 3x - 4$

5.  $y = 3x + 2$

## Part F: Slope Through Two Points

*Find the slope of the line through each pair of points.*

1.  $(-3, -2)$  and  $(0, 1)$

2.  $(-5, -10)$  and  $(-1, -2)$



3.  $(-1, -5)$  and  $(5, 13)$

4.  $(2, 3)$  and  $(4, 7)$

5.  $(-3, -8)$  and  $(-1, -4)$

### Part G: Equations From Two Points

*Write the equation of the line through each pair of points in slope-intercept form.*

1.  $(0, -2)$  and  $(3, 16)$

2.  $(-7, -58)$  and  $(8, 77)$

3.  $(-3, 2)$  and  $(4, 2)$

4.  $(7, 48)$  and  $(8, 56)$

5.  $(-4, -16)$  and  $(7, 6)$

### Part H: Equations From a Slope and a Point

*Write the equation of each line in slope-intercept form.*

1.  $m = 4$  through  $(-3, -17)$

2.  $m = 8$  through  $(0, -9)$

3.  $m = 4$  through  $(0, 4)$

4.  $m = \frac{7}{2}$  through  $(-2, 2)$

5.  $m = 4$  through  $(8, 34)$

**Part A: Negative Number Operations***Evaluate each expression completely.*

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*Solve each equation for  $x$ . Show your steps.*

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4.  $4x^2 \cdot x^2$



5.  $\frac{4x^6}{x^2}$

6.  $\frac{2x^8}{x^4}$

7.  $\frac{4x^{12}}{x^5}$

8.  $\frac{4x^7}{x^2}$

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## Part F: Slope Through Two Points

*Find the slope of the line through each pair of points.*

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2.  $(-5, -10)$  and  $(-1, -2)$

3.  $(-1, -5)$  and  $(5, 13)$

4.  $(2, 3)$  and  $(4, 7)$

5.  $(-3, -8)$  and  $(-1, -4)$

**Part G: Equations From Two Points**

*Write the equation of the line through each pair of points in slope-intercept form.*

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3.  $(-3, 2)$  and  $(4, 2)$

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5.  $(-4, -16)$  and  $(7, 6)$

**Part H: Equations From a Slope and a Point**

*Write the equation of each line in slope-intercept form.*

1.  $m = 4$  through  $(-3, -17)$

2.  $m = 8$  through  $(0, -9)$

3.  $m = 4$  through  $(0, 4)$

4.  $m = \frac{7}{2}$  through  $(-2, 2)$

5.  $m = 4$  through  $(8, 34)$

## Answer Key

### Part A: Negative Number Operations

- |          |           |            |
|----------|-----------|------------|
| 1. $-21$ | 8. $-6$   | 15. $-115$ |
| 2. $4$   | 9. $24$   | 16. $58$   |
| 3. $-8$  | 10. $18$  | 17. $63$   |
| 4. $-3$  | 11. $-11$ | 18. $-29$  |
| 5. $-5$  | 12. $-12$ | 19. $66$   |
| 6. $-16$ | 13. $72$  | 20. $-9$   |
| 7. $-5$  | 14. $-12$ |            |

### Part B: Solving Linear Equations

- |             |              |                        |
|-------------|--------------|------------------------|
| 1. $x = -9$ | 8. $x = -5$  | 15. $x = -9$           |
| 2. $x = -4$ | 9. $x = 4$   | 16. $x = -10$          |
| 3. $x = 5$  | 10. $x = -4$ | 17. $x = -10$          |
| 4. $x = 5$  | 11. $x = -1$ | 18. $x = \frac{11}{3}$ |
| 5. $x = -8$ | 12. $x = -7$ | 19. $x = -10$          |
| 6. $x = 6$  | 13. $x = 7$  | 20. $x = -3$           |
| 7. $x = 4$  | 14. $x = 7$  |                        |

### Part C: Linear Inequalities

- |                |                |                  |
|----------------|----------------|------------------|
| 1. $x \geq 11$ | 8. $x < 4$     | 15. $x < -8$     |
| 2. $x \geq 0$  | 9. $x > 2$     | 16. $x \geq -10$ |
| 3. $x \leq -5$ | 10. $x < -7$   | 17. $x < -5$     |
| 4. $x \leq 4$  | 11. $x \geq 6$ | 18. $x \leq -5$  |
| 5. $x > 3$     | 12. $x \geq 4$ | 19. $x \geq 10$  |
| 6. $x > -8$    | 13. $x \leq 8$ | 20. $x < -8$     |
| 7. $x \geq 11$ | 14. $x < 1$    |                  |

### Part D: Exponent Rules

- |              |                     |            |
|--------------|---------------------|------------|
| 1. $2x^{10}$ | 8. $4x^5$           | 15. $5000$ |
| 2. $6x^5$    | 9. $4x^{16}$        | 16. $14$   |
| 3. $2x^8$    | 10. $4x^{18}$       | 17. $343$  |
| 4. $4x^4$    | 11. $25x^4$         | 18. $9$    |
| 5. $4x^4$    | 12. $4096x^{16}$    | 19. $27$   |
| 6. $2x^4$    | 13. $\frac{7}{256}$ | 20. $49$   |
| 7. $4x^7$    | 14. $\frac{8}{81}$  |            |

### Part E: Slope and Linear Graphs

- |                          |                          |                         |
|--------------------------|--------------------------|-------------------------|
| 1. $m = 0, \quad b = -6$ | 3. $m = 2, \quad b = -3$ | 5. $m = 3, \quad b = 2$ |
| 2. $m = 1, \quad b = 6$  | 4. $m = 3, \quad b = -4$ |                         |

### Part F: Slope Through Two Points

1.  $m = 1$

2.  $m = 2$

3.  $m = 3$

4.  $m = 2$

5.  $m = 2$

**Part G: Equations From Two Points**

1.  $y = 6x - 2$

2.  $y = 9x + 5$

3.  $y = 2$

4.  $y = 8x - 8$

5.  $y = 2x - 8$

**Part H: Equations From a Slope and a Point**

1.  $y = 4x - 5$

2.  $y = 8x - 9$

3.  $y = 4x + 4$

4.  $y = \frac{7}{2}x + 9$

5.  $y = 4x + 2$